Considering a representative investor with wealth W_t in a 2-period economy. let $R_{f,t}$ be a baseline asset, we solve the following optimization problem:

$$\omega_t^* = \underset{\omega_t}{\text{argmax}}\ E_t[U(W_{t+1})] \quad \text{s.t.} \quad W_{t+1} = W_t \cdot R_{w,t+1} = W_t(R_{f,t+1} + \omega_t' \times R_{t+1})$$

in which $\times R_{t+1}$ is the vector of asset excess return relative to $R_{f,t}$

Using the power utility function $U(W) = \frac{W^{1-\gamma}}{1-\gamma}$ where γ is CRRA, we rewrite the problem as:

$$\omega_t^* = \underset{\omega_t}{\text{argmax}}\ E_t\left[\frac{\left(W_t \cdot (R_{f,t+1} + \omega_t' \times R_{t+1})\right)^{1-\gamma}}{1-\gamma}\right]$$

Thus, first order condition of the maximization problem is:

$$E_t\left[\frac{(1-\gamma)\left[W_t \cdot (R_{f,t+1} + \omega_t' \times R_{t+1})\right]^{-\gamma} \cdot \times R_{t+1}}{1-\gamma}\right] = 0$$

$$\Rightarrow \quad E_t\left[\left(R_{f,t+1} + \omega_t' \times R_{t+1}\right)^{-\gamma} \cdot \times R_{t+1}\right] = 0$$

$$\Rightarrow \quad E_t\left[\left(R_{w,t+1}\right)^{-\gamma} \cdot \times R_{t+1}\right] = 0$$

As a result, the SDF $M_t = R_{w,t+1}^{-\gamma}$ prices the vector of excess returns.